Reversible reactions

SOME REACTIONS CAN BE REVERSED.

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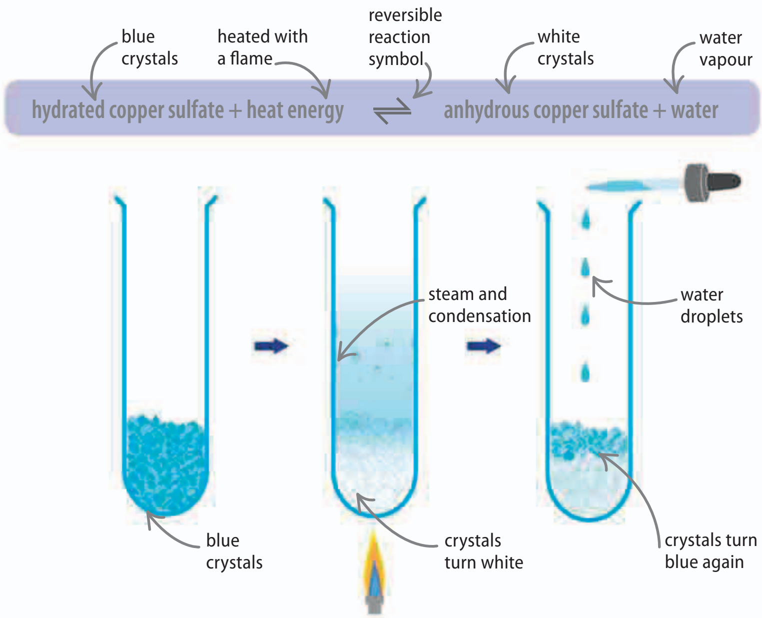
In general, chemical reactions run in just one direction. The energy that is required to turn the products back into reactants is just too great for it to happen. However, some reactions are easily reversible.

Two-way reactions

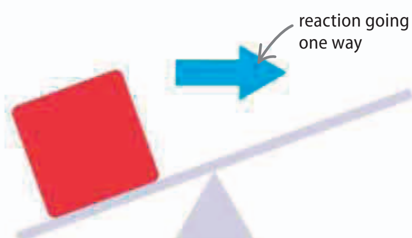
A reversible reaction is one that can go backward as well as forward—products that form can easily be turned back into the original reactants. The amounts of energy needed to make the reaction run in either direction are rarely equal, but there is not a large difference between the two. A common reversible reaction is to use heat to drive water from a solid. This is reversed by simply adding water.



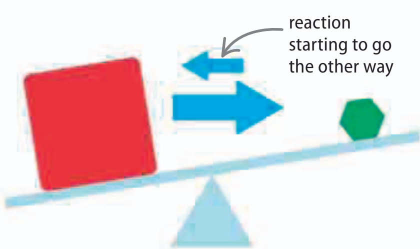
▷ **Hydrating crystals**
Copper sulfate crystals are blue due to water molecules locked inside them. Heating the crystals drives out water and they turn into the white anhydrous (without water) form. However, adding water easily reverses the process.

Dynamic equilibrium

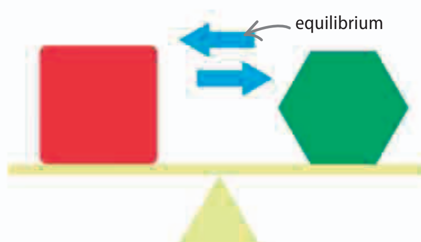
Reversible reactions do not normally run one way and then the other. They run in both directions simultaneously. However, it is the rate of reaction in each direction that dictates the yield (the proportions of reactants and products). When the rate of reaction in both directions is the same, the reaction is in equilibrium.



△ **One way**
At the start of the reaction, few products have formed so the reaction runs in one direction. The high concentration of reactants makes the rate of reaction high.



△ **Mostly one direction**
Although there is still a high concentration of reactants, the reaction is starting to go backward. However, the concentration of products is increasing.



△ **Dynamic equilibrium**
Products are being made at the same speed as they turn back into reactants but the reactions continue. This stage in the reaction is called dynamic equilibrium.